Introduction to Internet of Things

Assignment-Week 2

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15 **Total marks: 15 X 1= 15**

QUESTION 1:

Based on functionality, MQTT is a _____ protocol.

a. Transport
b. Data
c. Semantic
d. None of these

Correct Answer: b. Data

Detailed Solution: MQTT is a Data Protocol.

See lecture 6 (Basics of IoT Networking – Part II) @ 01:30

QUESTION 2:

MQTT is a _____-based lightweight protocol for use in conjunction with TCP/IP protocol.

a. Client-Server
b. Publish-Subscribe
c. Both (a) and (b)
d. Neither (a) nor (b)

Correct Answer: b. Publish-Subscribe

Detailed Solution: MQTT, is a publish-subscribe-based lightweight protocol for use in conjunction with TCP/IP protocol.

See lecture 6 (Basics of IoT Networking – Part II) @ 02:12

QUESTION 3:

A _____ controls the publish-subscribe messaging pattern in MQTT.

a. Publishers
b. Subscribers
c. Broker
d. None of these

Correct Answer: c. Broker

Detailed Solution: A Broker controls the publish-subscribe messaging pattern in MQTT.

See lecture 6 (Basics of IoT Networking – Part II) @ 03:31

QUESTION 4:

State True or False.

Statement: "In MQTT, the Publishers are Lightweight Sensors."

a. True
b. False

Correct Answer: a. True

Detailed Solution: In MQTT, the Publishers are lightweight sensors.

See lecture 6 (Basics of IoT Networking – Part II) @ 04:49

QUESTION 5:

Which of the following is MQTT designed for?

a. Remote connections
b. Limited bandwidth
c. Small code footprint
d. All of these

Correct Answer: d. All of these

Detailed Solution: MQTT is designed for –

a. Remote connections
b. Limited bandwidth
c. Small code footprint

See lecture 6 (Basics of IoT Networking – Part II) @ 03:23

QUESTION 6:

A _____ to which a client is subscribed is updated in the form of messages and distributed by the _____.

a. Topic, Message Broker
b. Message Broker, Topic
c. Message Broker, Message Broker
d. Topic, Topic

Correct Answer: a. Topic, Message Broker

Detailed Solution: A topic to which a client is subscribed is updated in the form of messages and distributed by the message broker.

See lecture 6 (Basics of IoT Networking – Part II) @ 04:00

QUESTION 7:

There are _____ number of methods in MQTT protocol.

a. 5
b. 4
c. 6
d. None of these

Correct Answer: a. 5

Detailed Solution: There are 5 number of methods in MQTT protocol.

See lecture 6 (Basics of IoT Networking – Part II) @ 05:49

QUESTION 8:

State True or False.

Statement: "Publish/Subscribe in MQTT is event-driven and enables messages to be pushed to clients."

a. True
b. False

Correct Answer: a. True

Detailed Solution: Publish/Subscribe in MQTT is event-driven and enables messages to be pushed to clients.

See lecture 6 (Basics of IoT Networking – Part II) @ 08:32

QUESTION 9:

SMQTT stands for?

a. Secure MQTT
b. Safe MQTT
c. Safety in MQTT
d. Stored MQTT

Correct Answer: a. Secure MQTT

Detailed Solution: SMQTT stands for Secure MQTT.

See lecture 6 (Basics of IoT Networking – Part II) @ 14:10

QUESTION 10:

Similar to HTTP, CoAP utilizes which of the following for operation?

a. GET
b. PUT
c. PUSH
d. All of these

Correct Answer: d. All of these

Detailed Solution: Similar to HTTP, CoAP utilizes GET, PUT, PUSH, DELETE message requests to retrieve, create, update, and delete, respectively.

See lecture 7 (Basics of IoT Networking – Part III) @ 08:24

QUESTION 11:

State True or False.

Statement: "The CoAP protocol is a Client-Server interaction-based protocol that is asynchronous over a datagram-oriented transport protocol such as UDP."

a. True
b. False

Correct Answer: a. True

Detailed Solution: The CoAP protocol is a Client-Server interaction-based protocol that is asynchronous over a datagram-oriented transport protocol such as UDP.

See lecture 7 (Basics of IoT Networking – Part III) @ 00:30

QUESTION 12:

What is the full form of M2M in the context of IoT protocol?

a. Message to Message
b. Man to Message
c. Message to Man
d. None of these

Correct Answer: d. None of these

Detailed Solution: M2M stands for Machine to Machine.

See lecture 7 (Basics of IoT Networking – Part IV) @ 0:30

QUESTION 13:

What is the basic unit of data in the AMQP protocol?

a. Chunk
b. Byte
c. Slit
d. None of these

Correct Answer: d. None of these

Detailed Solution: In AMQP, the basic unit of data is called a frame.

See lecture 8 (Basics of IoT Networking – Part IV) @ 0:41

QUESTION 14:

State True or False.

Statement: "The OSI model has 9 layers."

a. True
b. False

Correct Answer: b. False

Detailed Solution: The OSI model is a conceptual framework that divides any networked communication system into seven layers.

See Page number – 10, Chapter - 1, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)

QUESTION 15:

The “Destination Address” in the IPv4 packet represents which of the following?

a. The destination node address of the packet
b. The intermediate hop in the network
c. Both (a) and (b)
d. Neither (a) nor (b)

Correct Answer: a. The destination node address of the packet

Detailed Solution: The “Destination Address” in the IPv4 packet represents the address of the destination node in the network.

See Page number – 18, Chapter - 1, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)

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